

VLAN Trunking & Router-on-a-Stick Lab

Topology

PC01 --- SW0 --- SW1 --- PC03
PC02 --- SW0 SW1 --- PC04
|
Router1

VLAN Assignment

Device	VLAN	IP Address
PC01	10	192.168.10.10/24
PC03	10	192.168.10.20/24
PC02	20	192.168.20.10/24
PC04	20	192.168.20.20/24

Default Gateways

- VLAN 10 → 192.168.10.1
- VLAN 20 → 192.168.20.1

Switch0 Configuration

```
enable
configure terminal

vlan 10
  name SALES

vlan 20
  name ENG

interface fa0/1
  switchport mode access
  switchport access vlan 10

interface fa0/2
  switchport mode access
  switchport access vlan 20

interface gi0/1
  switchport mode trunk

interface gi0/2
  switchport mode trunk

end
copy running-config startup-config
```

Switch1 Configuration

```
enable
configure terminal

vlan 10
  name SALES

vlan 20
  name ENG

interface fa0/1
  switchport mode access
  switchport access vlan 10

interface fa0/2
  switchport mode access
  switchport access vlan 20

interface gi0/1
  switchport mode trunk

end
copy running-config startup-config
```

Router1 Configuration (Router-on-a-Stick)

```
enable
configure terminal

interface g0/0/0
  no shutdown

interface g0/0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0

interface g0/0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0

end
copy running-config startup-config
```

Verification Commands

Check VLANs

```
show vlan brief
```

Check Trunk Ports

```
show interfaces trunk
```

Check Router Interfaces

```
show ip interface brief
```

Expected:

```
Gi0/0/0      up/up
```

Gi0/0/0.10	up/up
Gi0/0/0.20	up/up

Connectivity Tests

From PC01:

```
ping 192.168.10.1
ping 192.168.10.20
ping 192.168.20.10
```

From PC02:

```
ping 192.168.20.1
ping 192.168.20.20
ping 192.168.10.10
```

Key Concepts

Access Port:

- Carries one VLAN
- Used for PCs

Trunk Port:

- Carries multiple VLANs
- Uses 802.1Q tagging

Router-on-a-Stick:

- One physical router interface
- Multiple subinterfaces
- Each subinterface serves one VLAN gateway

Traffic Flow:

PC → Access Port → VLAN → Trunk → Router →
Trunk → VLAN → Destination